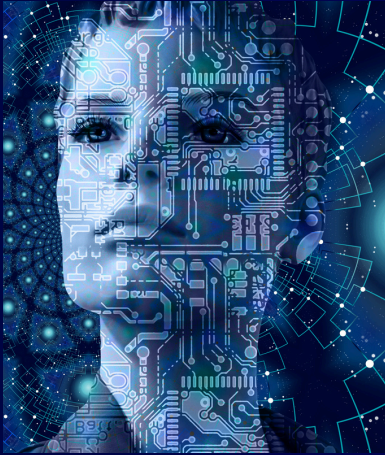


DATA ANALYSIS PROJECTS

JAI KHANUJA

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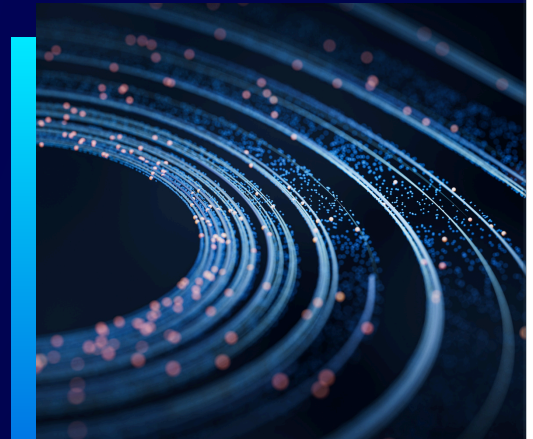


THE IMPORTANCE OF DATA ANALYSIS

In today's digital age, data is a critical asset. Effective data analysis enables organizations to identify trends, improve operations, and discover new opportunities. It bridges the gap between data collection and strategic implementation.

PROJECT

E-COMMERCE FURNITURE SALES DATASET 2024
ELECTRIC VEHICLE SALES BY STATE IN INDIA
GOOGLE PLAY STORE APPS
OLYMPIC SUMMER MEDAL FROM 1976-2008



E COMMERCE FURNITURE SALES DATA SET SUMMARY

PROJECT OVERVIEW

TITLE: E-COMMERCE FURNITURE SALES

DOMAIN: DATA ANALYST

**TOOL USED: PYTHON, PANDAS, SCIKIT-LEARN, MATPLOTLIB, SEABORN,
EXCEL**

**DATASET: CONTAINS 2,000 ALIEXPRESS FURNITURE PRODUCT ENTRIES WITH
SALES AND PRODUCT DETAILS.**

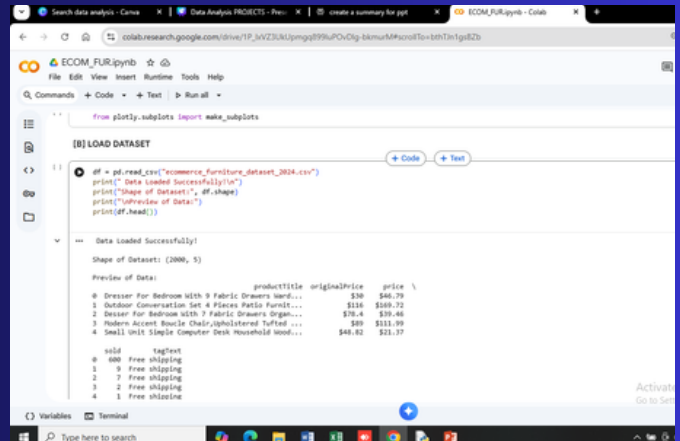
PROJECT OBJECTIVE

Main Goal: Predict the number of furniture items sold based on features like product title, original price, current price, and tags.

DATASET DETAILS

FIELDS INCLUDE:

- PRODUCTTITLE
 - ORIGINALPRICE
 - PRICE
 - SOLD
 - TAGTEXT (E.G., "FREE SHIPPING").
- USEFUL FOR: EXPLORATORY DATA ANALYSIS, SALES PREDICTION, MARKET TRENDS, IMPACT OF DISCOUNTS, PRODUCT POPULARITY ANALYSIS.
- DATA COLLECTED ETHICALLY FROM ALIEXPRESS



The screenshot shows a Jupyter Notebook interface with a code cell titled "[0] LOAD DATASET". The code reads a CSV file from Google Drive, prints the shape of the dataset, and displays a preview of the data. The output shows the dataset has a shape of (2000, 5) and a preview of the first five rows.

```
from google.colab import drive
drive.mount('/content/drive')

df = pd.read_csv('ecommerce_furniture_dataset_2024.csv')
print("Data loaded successfully!")
print("Shape of Dataset:", df.shape)
print("Preview of Data:")
print(df.head())
```

Data loaded successfully!

Shape of Dataset: (2000, 5)

Preview of Data:

	producttitle	originalPrice	price	
0	Dresser For Bedroom With 9 Fabric Drawers Ward...	\$130	\$40.79	
1	Outdoor Conversation Set & Flame Patio Furnit...	\$110	\$100.72	
2	Dresser For Bedroom With 7 Fabric Drawers Organ...	\$70.4	\$39.46	
3	Modern Accent Boucle Chair Upholstered Tufted ...	\$69	\$111.99	
4	Small Unit Simple Computer Desk Household Wood...	\$40.02	\$21.77	

Variables Terminal

METHODOLOGY

1. **Data Collection:** Dataset loaded via pandas as a CSV file.
2. **Data Preprocessing:**
 - **Handling missing values** (dropping incomplete rows)
 - **Encoding categorical variables** (tagText and productTitle via LabelEncoder and TF-IDF)
 - **Feature engineering** (e.g., calculating percentage discount).

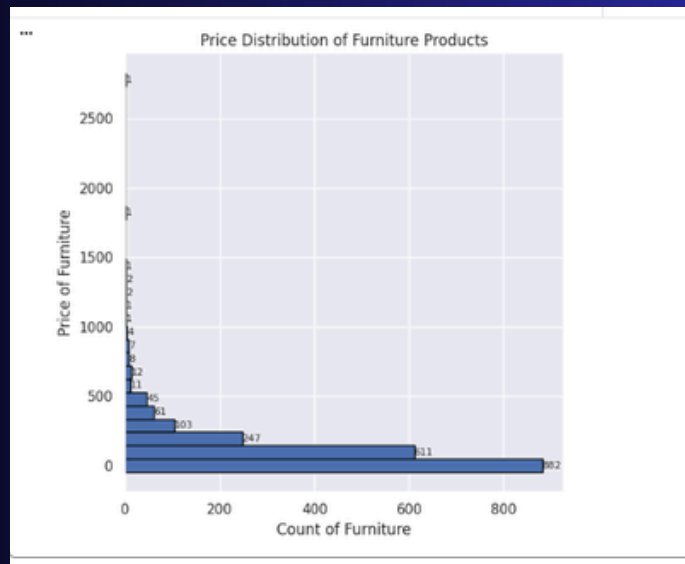
✓ Cleaned Data Sample:

	productTitle	originalPrice	price	\
0	Dresser For Bedroom With 9 Fabric Drawers Ward...	30.00	46.79	
1	Outdoor Conversation Set 4 Pieces Patio Furnit...	116.00	169.72	
2	Desser For Bedroom With 7 Fabric Drawers Organ...	78.40	39.46	
3	Modern Accent Boucle Chair,Upholstered Tufted ...	89.00	111.99	
4	Small Unit Simple Computer Desk Household Wood...	48.82	21.37	

	sold	tagText
0	600	Free shipping
1	9	Free shipping
2	7	Free shipping
3	2	Free shipping
4	1	Free shipping

EXPLORATORY DATA ANALYSIS

PRICE DISTRIBUTION OF FURNITURE



DISTRIBUTION OF ITEMS SOLD



PRICE VS SOLD FURNITURE



SHIPPING OPTION DISTRIBUTION



FEATURE ENGINEERING

```
-- 10287 features added and 'productTitle' dropped.

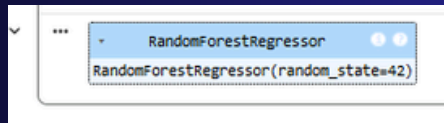
Feature engineering completed!
originalPrice price sold tagText discount_percentage adjustable and \
0 38.0 46.79 680 99 -55.966667 0.0 0.0
1 111.0 169.72 9 99 -41.189545 0.0 0.0
2 78.4 39.46 7 99 -49.663637 0.0 0.0

bed bedroom bedside ... up upholstered vanity velvet wardrobe \
0 0.0 0.189721 0.0 ... 0.0 0.0 0.0 0.3006111
1 0.0 0.200000 0.0 ... 0.0 0.0 0.0 0.0000000
2 0.0 0.200469 0.0 ... 0.0 0.0 0.0 0.0000000

white vicker with wood wooden
0 0.0 0.000000 0.128182 0.0 0.0
1 0.0 0.200000 0.000000 0.0 0.0
2 0.0 0.000000 0.118991 0.0 0.0

[3 rows x 145 columns]
```

MODEL TRAINING PREDICT NUMBER OF ITEMS SOLD



MODEL EVALUTION

✓ ☒ MODEL PERFORMANCE:
Linear Regression → MSE: 1108199.66, R^2 : -0.208
Random Forest → MSE: 935688.42, R^2 : -0.020

KEY INSIGHT



INSIGHTS:

- 1 Total Products: 538
- 2 Average Price: \$136.75
- 3 Average Discount: -5428.00%
- 4 Most Common Shipping Type: 99

EV SALES BY STATES INDIA

SUMMARY

PROJECT OVERVIEW

TITLE: EV SALES BY STATE IN INDIA

DOMAIN: DATA ANALYST

**TOOL USED: PYTHON, PANDAS, SCIKIT-LEARN, MATPLOTLIB, SEABORN,
EXCEL**

**DATASET: ALL THE TYPES OF ELECTRICAL VEHICLE SOLD ALL OVER THE
STATES OF INDIA**

PROJECT OBJECTIVE

The goal of the "Electric Vehicle Sales by State in India" project is to analyze and predict electric vehicle (EV) sales trends across different states in India, providing insights into regional adoption patterns, market drivers, and key factors affecting EV uptake. The project aims to help guide strategic decisions for market planning, investment, and infrastructure development in the growing Indian EV sector.

DATASET DETAILS

```
df = pd.read_csv("Electric-Vehicle-Sales-by-State-in-India.csv",
                 engine='python',      # allows advanced parsing
                 on_bad_lines='skip',   # skips lines with parsing
                                     errors
                 encoding_errors='ignore') # ignores text encoding
                                     errors

# Confirmation message
print(" Data Loaded Successfully!\n")

# Show the shape (rows, columns)
print("Shape of dataset:", df.shape)

# Show the first 5 rows
print("\nPreview:")
print(df.head())
```

```
... Data Loaded Successfully!

Shape of dataset: (96845, 8)

Preview:
  Year Month_Name  Date  State  Vehicle_Class \
0  2014      jan  01-01-2014  Andhra Pradesh  ADAPTED VEHICLE
1  2014      jan  01-01-2014  Andhra Pradesh  AGRICULTURAL TRACTOR
2  2014      jan  01-01-2014  Andhra Pradesh  AMBULANCE
3  2014      jan  01-01-2014  Andhra Pradesh  ARTICULATED VEHICLE
4  2014      jan  01-01-2014  Andhra Pradesh  BUS

  Vehicle_Category  Vehicle_Type  EV_Sales_Quantity
0      Others      Others                2
1      Others      Others                6
2      Others      Others                9
3      Others      Others                2
4      Bus         Bus                 33
```


METHODOLOGY

1. Data Collection: Loaded CSV into pandas; inspected shape and types.
2. Preprocessing:
 - Converted data types
 - Encoded categorical variables
 - Created new features for time analysis (month/day)

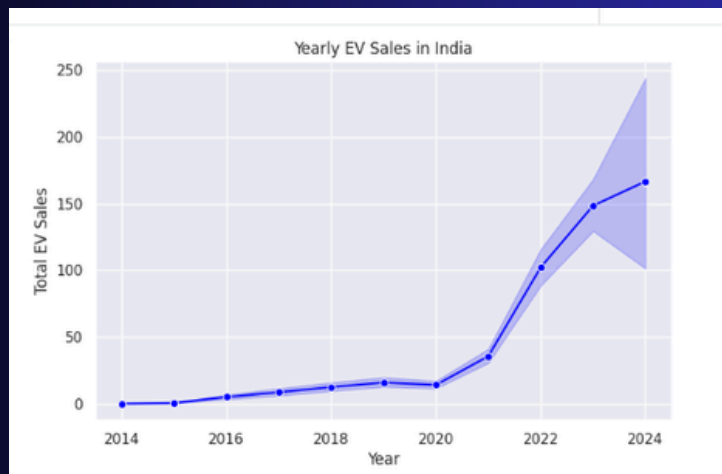
```
***
Cleaned Data Info:
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 96845 entries, 0 to 96844
Data columns (total 8 columns):
 #   Column              Non-Null Count  Dtype
---  --
 0   Year                96845 non-null  int64
 1   Month_Name          96845 non-null  category
 2   Date                96845 non-null  datetime64[ns]
 3   State               96845 non-null  category
 4   Vehicle_Class       96845 non-null  category
 5   Vehicle_Category    96845 non-null  category
 6   Vehicle_Type        96845 non-null  category
 7   EV_Sales_Quantity   96845 non-null  int64
dtypes: int64(1), category(5), datetime64[ns](1), int64(1)
memory usage: 2.8 MB
None

Missing Values Summary:
Year                0
Month_Name          0
Date                0
State               0
Vehicle_Class       0
Vehicle_Category    0
Vehicle_Type        0
EV_Sales_Quantity   0
dtype: int64

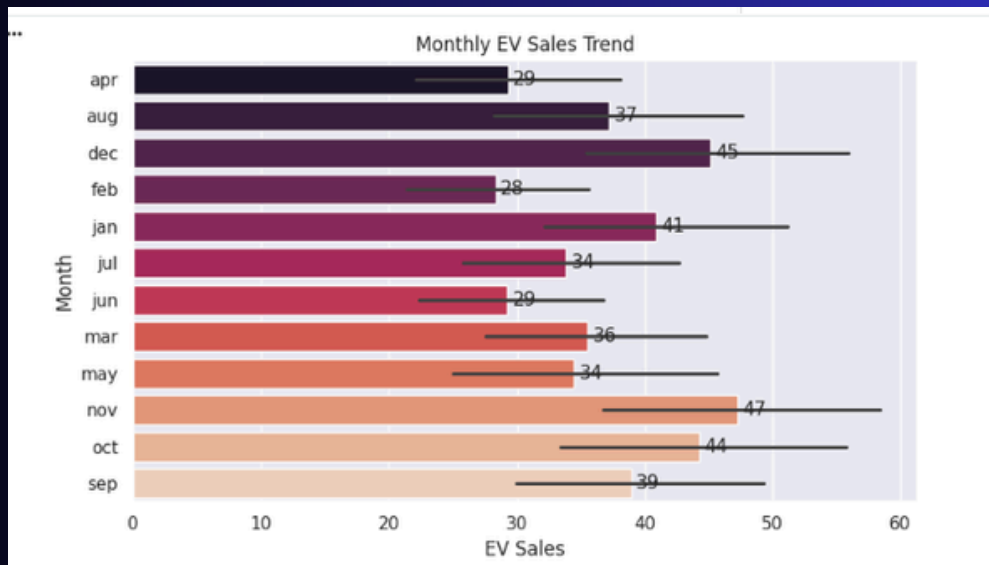
Statistical Summary (Numeric Columns):
      Year                Date  EV_Sales_Quantity
count  96845.0            96845            96845.000000
mean  2015.622768  2019-01-28 11:00:33.641385728            37.125933
min    2014.0            2014-01-01 00:00:00            0.000000
25%    2016.0            2016-06-01 00:00:00            0.000000
50%    2019.0            2019-07-01 00:00:00            0.000000
```

DATA VISUALIZTIONS

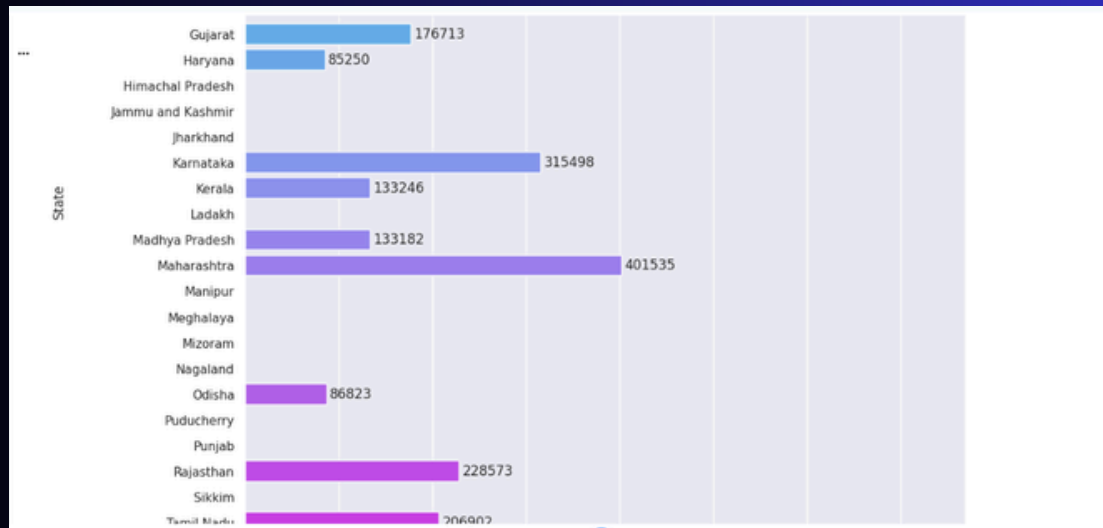
YEARLY TREND OF EV SALES



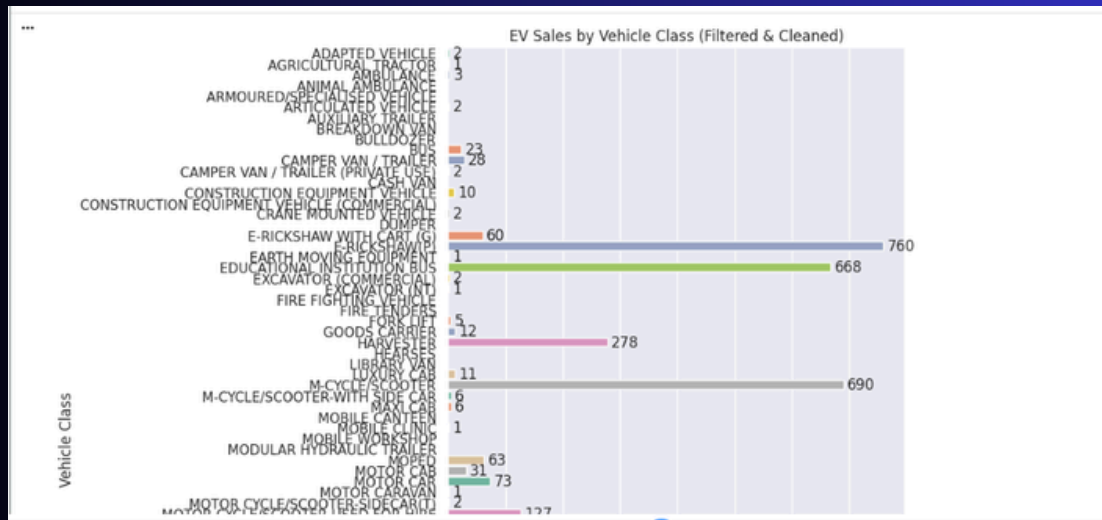
MONTHLY EV SALES TREND



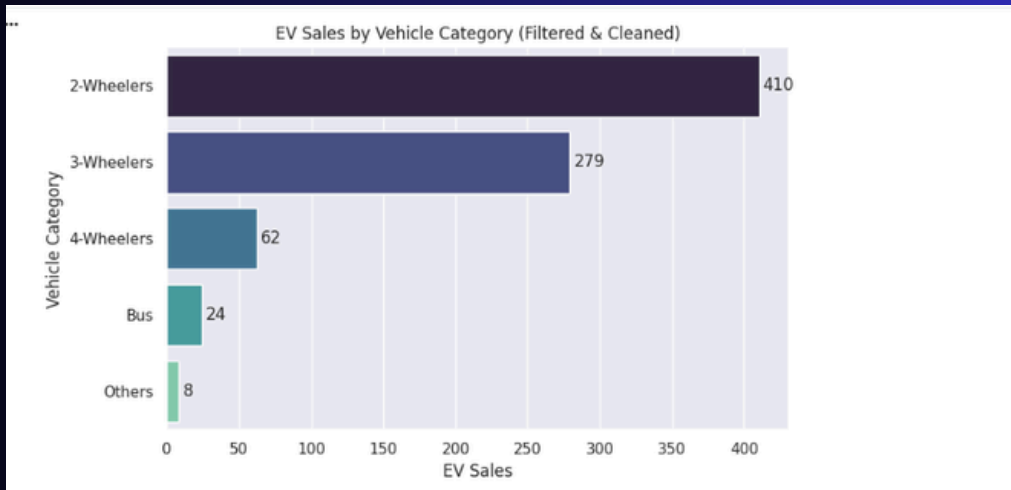
STATE WISE EV SALES



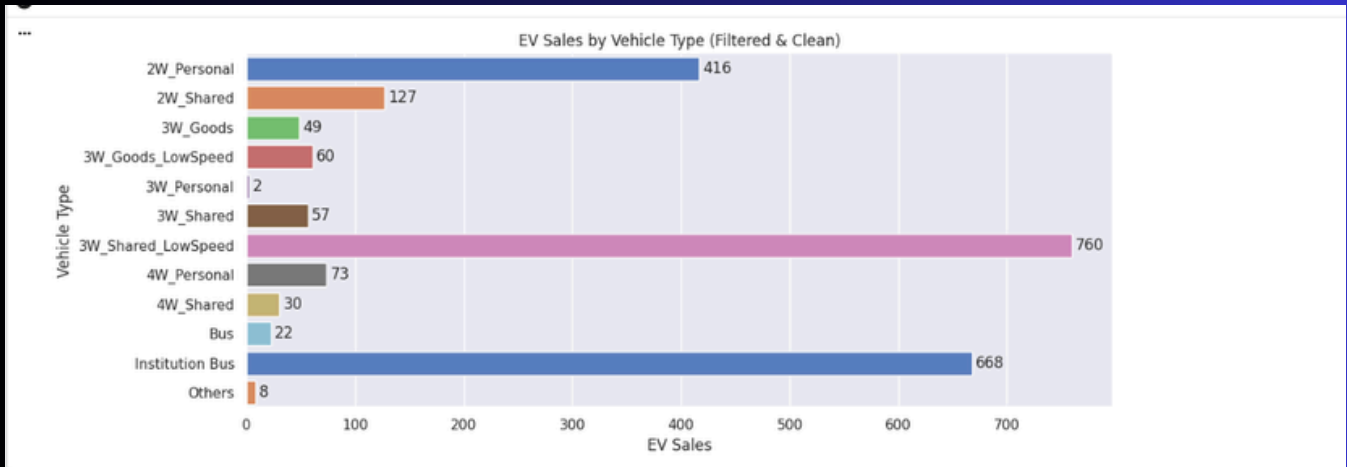
EV SALES BY VEHICLE CLASS



EV SALES BY VEHICLE CATEGORY



EV SALES BY VEHICLE TYPE



KEY INSIGHTS

INSIGHTS:

- 1 Total Records: 96845
- 2 Time Range: 2014 to 2024
- 3 Highest Selling State: Uttar Pradesh (732074 sales)
- 4 Highest Vehicle Category: 2-Wheelers (1808105 sales)
- 5 Average EV Sales Quantity: 37.13

GOOGLE PLAY STORE DATA ANALYSIS

SUMMARY

PROJECT OVERVIEW

TITLE: GOOGLE PLAY STORE DATA ANALYSIS

DOMAIN: DATA ANALYST

**TOOL USED: PYTHON, PANDAS, SCIKIT-LEARN, MATPLOTLIB, SEABORN,
EXCEL**

**THE DATASET HAS 10,800+ ROWS AND 13 MAIN COLUMNS SUCH AS APP,
CATEGORY, RATING, REVIEWS, SIZE, INSTALLS, TYPE, PRICE, CONTENT RATING,
GENRES, LAST UPDATED, CURRENT VER, AND ANDROID VER.**

PROJECT OBJECTIVE

The objective is to analyze app characteristics (ratings, reviews, size, installs, price, etc.) to find trends, outliers, and patterns that can guide app-business decisions.

DATASET DETAILS

```
apps_df=pd.read_csv("googleplaystore.csv")
reviews_df=pd.read_csv("googleplaystore_user_reviews.csv")
print("App Dataset Shape:",apps_df.shape)
print("User Reviews Dataset Shape:",reviews_df.shape)
apps_df.head()
```

App Dataset Shape: (10041, 13) User Reviews Dataset Shape: (64295, 5)													
	App	Category	Rating	Reviews	Size	Installs	Type	Price	Content Rating	Genres	Last Updated	Current Ver	Android Ver
0	Photo Editor & Candy Camera & Grid & ScrapBook	ART_AND_DESIGN	4.1	159	19M	10,000+	Free	0	Everyone	Art & Design	January 7, 2018	1.0.0	4.0.3 and up
1	Coloring book moana	ART_AND_DESIGN	3.9	967	14M	500,000+	Free	0	Everyone	Art & Design;Pretend Play	January 15, 2018	2.0.0	4.0.3 and up
2	U Launcher Lite – FREE Live Cool Themes, Hide ...	ART_AND_DESIGN	4.7	87510	8.7M	5,000,000+	Free	0	Everyone	Art & Design	August 1, 2018	1.2.4	4.0.3 and up
3	Sketch - Draw & Paint	ART_AND_DESIGN	4.5	215644	25M	50,000,000+	Free	0	Teen	Art & Design	June 8, 2018	Varies with device	4.2 and up
4	Pixel Draw - Number Art Coloring Book	ART_AND_DESIGN	4.3	967	2.8M	100,000+	Free	0	Everyone	Art & Design;Creativity	June 20, 2018	1.1	4.4 and up

METHODOLOGY

EXPLORE BASIC INFO

```
Dataset Information
...
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 10041 entries, 0 to 10040
Data columns (total 13 columns):
#   Column              Non-Null Count  Dtype
---  -
0   App                  10041 non-null  object
1   Category             10041 non-null  object
2   Rating               9367 non-null   float64
3   Reviews              10041 non-null  object
4   Size                 10041 non-null  object
5   Installs              10041 non-null  object
6   Type                 10040 non-null  object
7   Price                10041 non-null  object
8   Content Rating       10040 non-null  object
9   Genres               10041 non-null  object
10  Last Updated         10041 non-null  object
11  Current Ver          10033 non-null  object
12  Android Ver          10038 non-null  object
dtypes: float64(1), object(12)
memory usage: 1.1+ MB
None

Missing Values
App                0
Category           0
Rating             1474
Reviews            0
Size               0
Installs           0
Type               1
Price              0
Content Rating     1
Genres             0
Last Updated       0
```

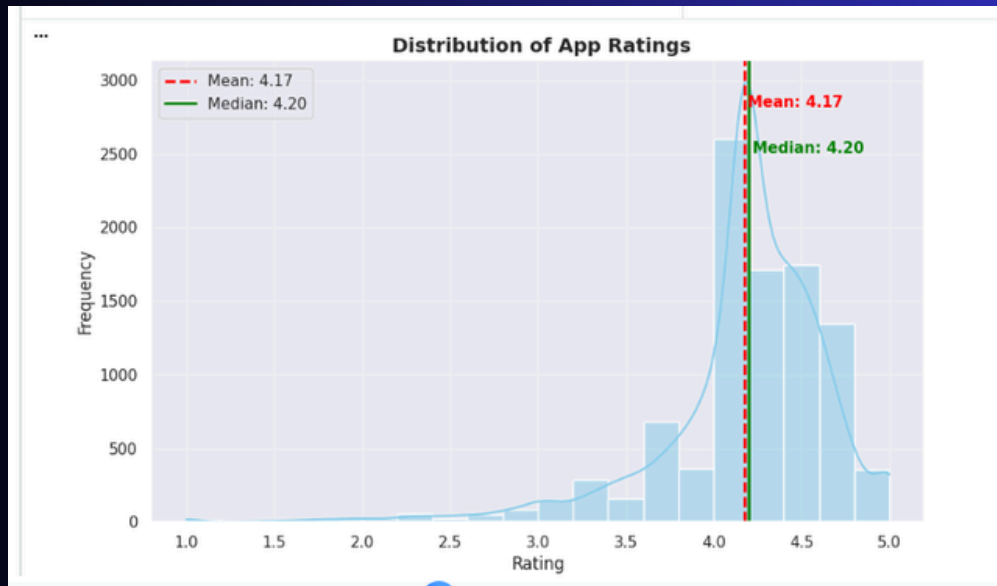
SUMMARY STATISTICS

Summary Of Numeric Columns:

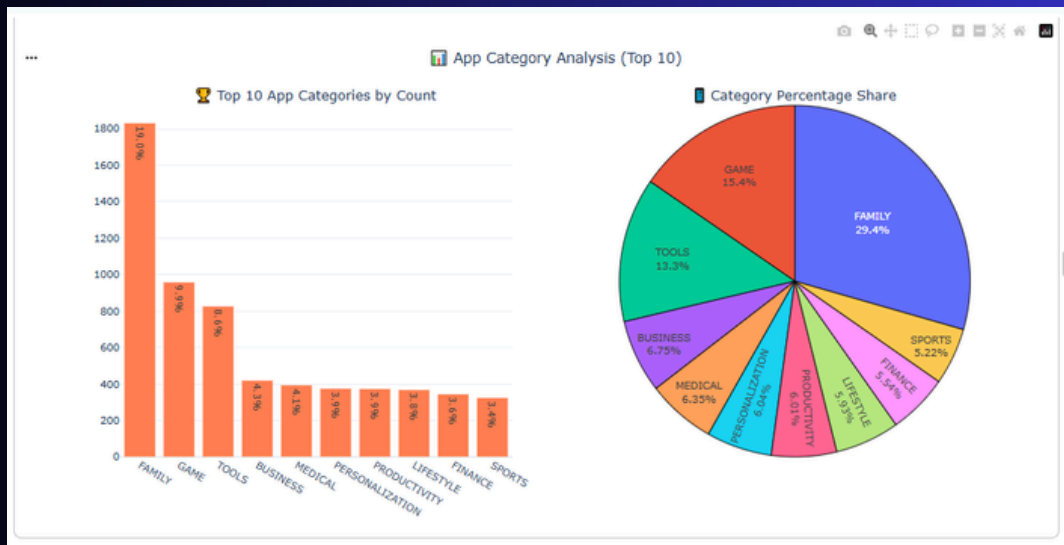
	Rating	Reviews	Installs	Price	Size_KB
count	9659.000000	9.659000e+03	9.659000e+03	9659.000000	8432.000000
mean	4.173517	2.165926e+05	7.777507e+06	1.099299	20884.374134
std	0.494313	1.831320e+06	5.375828e+07	16.852152	22351.767303
min	1.000000	0.000000e+00	0.000000e+00	0.000000	8.500000
25%	4.000000	2.500000e+01	1.000000e+03	0.000000	4710.400000
50%	4.200000	9.670000e+02	1.000000e+05	0.000000	12288.000000
75%	4.500000	2.940100e+04	1.000000e+06	0.000000	28672.000000
max	5.000000	7.815831e+07	1.000000e+09	400.000000	102400.000000

DATA VISUALIZTIONS

DISTRIBUTION OF APPS RATING



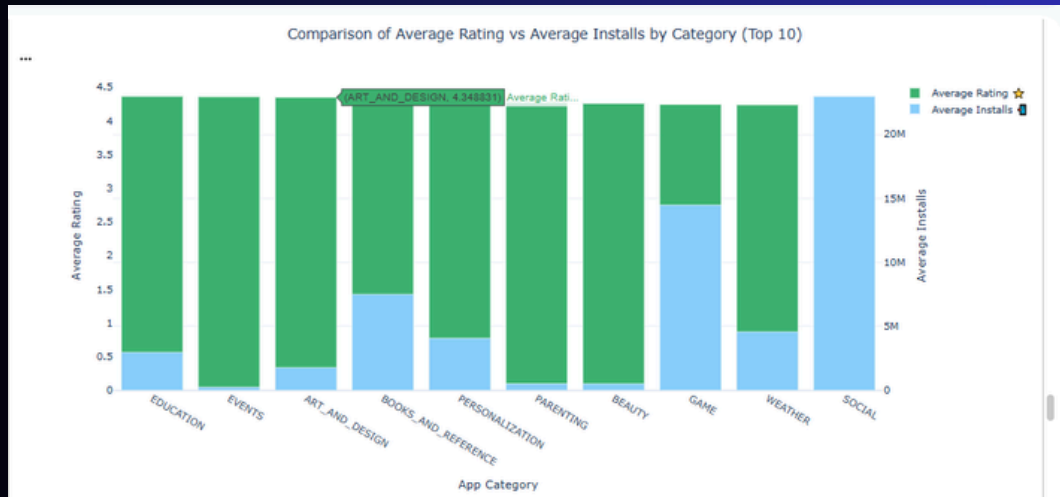
APP CATEGORIES (TOP 10)



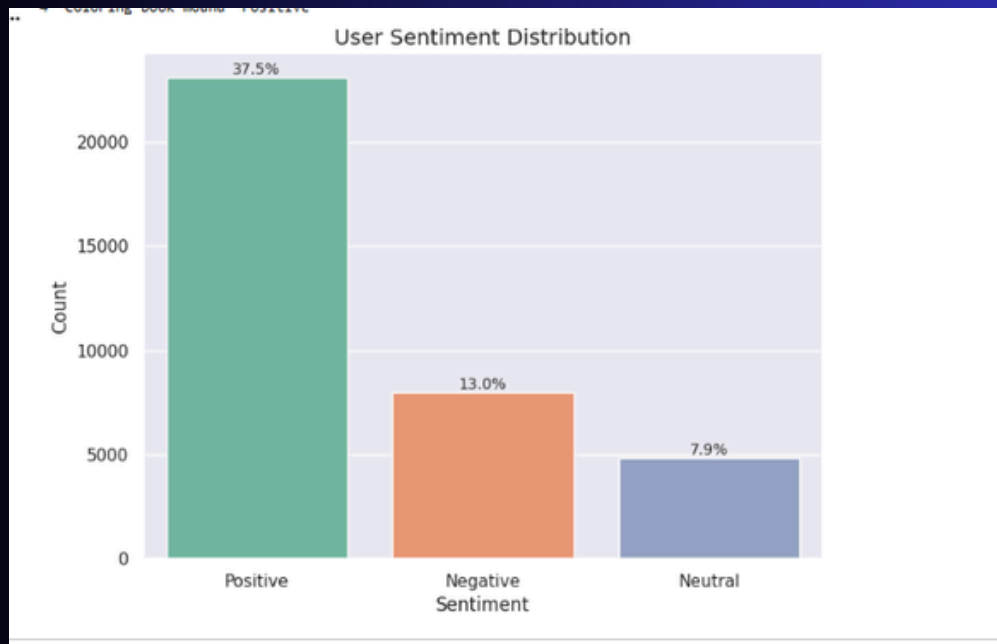
RATING VS INSTALL



AVERAGE RATING



THE USER REVIEW



KEY INSIGHTS

...	App Dataset — Key Insights	
	Metric	Value
	Total Apps Analyzed	9,659
	Average App Rating	4.170000
	Median App Rating	4.200000
	Most Popular Category	FAMILY
	Most Installed App	Google Play Books
	Most Installed App (Installs)	1,000,000,000
	Total Installs (All Apps)	75,122,937,527

SUMMER OLYMPIC 1976- 2008

SUMMARY

PROJECT OVERVIEW

TITLE: SUMMER OLYMPIC 1976-2008

DOMAIN: DATA ANALYST

**TOOL USED: PYTHON, PANDAS, SCIKIT-LEARN, MATPLOTLIB, SEABORN,
EXCEL**

PROJECT OBJECTIVE

Analyze Olympic medal trends from 1976–2008.
Identify top-performing countries, athletes, and sports.
Explore gender participation patterns.
Build a machine learning model to predict medal outcomes based on attribute

DATASET DETAIL

```
data=pd.read_csv("Summer-Olympic-medals-1976-  
to-2008.csv";encoding="latin1")  
print("DATASET LOADED SUCCESSFULLY ")  
print("SHAPE OF DATASET",data.shape)  
print("\nPreview:")  
print(data.head())
```

```
*** DATASET LOADED SUCCESSFULLY  
SHAPE OF DATASET (15433, 12)  
  
Preview:  
   City  Year  Sport Discipline  Event \  
0  Montreal  1976.0  Aquatics  Diving  3m springboard  
1  Montreal  1976.0  Aquatics  Diving  3m springboard  
2  Montreal  1976.0  Aquatics  Diving  3m springboard  
3  Montreal  1976.0  Aquatics  Diving  3m springboard  
4  Montreal  1976.0  Aquatics  Diving  10m platform  
  
   Athlete  Gender  Country_Code  Country  Event_gender \  
0  KÖHLER, Christa  Women  GDR  East Germany  W  
1  KOSENKOV, Aleksandr  Men  URS  Soviet Union  M  
2  BOGGS, Philip George  Men  USA  United States  M  
3  CAGNOTTO, Giorgio Franco  Men  ITA  Italy  M  
4  WILSON, Deborah Keplar  Women  USA  United States  W  
  
   Medal  Count  
0  Silver  NaN  
1  Bronze  NaN  
2  Gold  NaN  
3  Silver  NaN  
4  Bronze  NaN
```

Methodology

DATA CLEANING

*** Checking for missing values:

```
City          117
Year          117
Sport         117
Discipline    117
Event         117
Athlete       117
Gender        117
Country_Code  117
Country       117
Event_gender  117
Medal         117
Count        15433
dtype: int64
```

✅ Cleaned Data:

```
<class 'pandas.core.frame.DataFrame'>
Index: 15316 entries, 0 to 15432
Data columns (total 10 columns):
#   Column      Non-Null Count  Dtype
---  -
0   City        15316 non-null  object
1   Year        15316 non-null  int64
2   Sport       15316 non-null  object
3   Discipline  15316 non-null  object
4   Event       15316 non-null  object
5   Athlete     15316 non-null  object
6   Gender      15316 non-null  object
7   Country     15316 non-null  object
8   Medal       15316 non-null  object
9   Count       0 non-null      float64
dtypes: float64(1), int64(1), object(8)
memory usage: 1.3+ MB
```

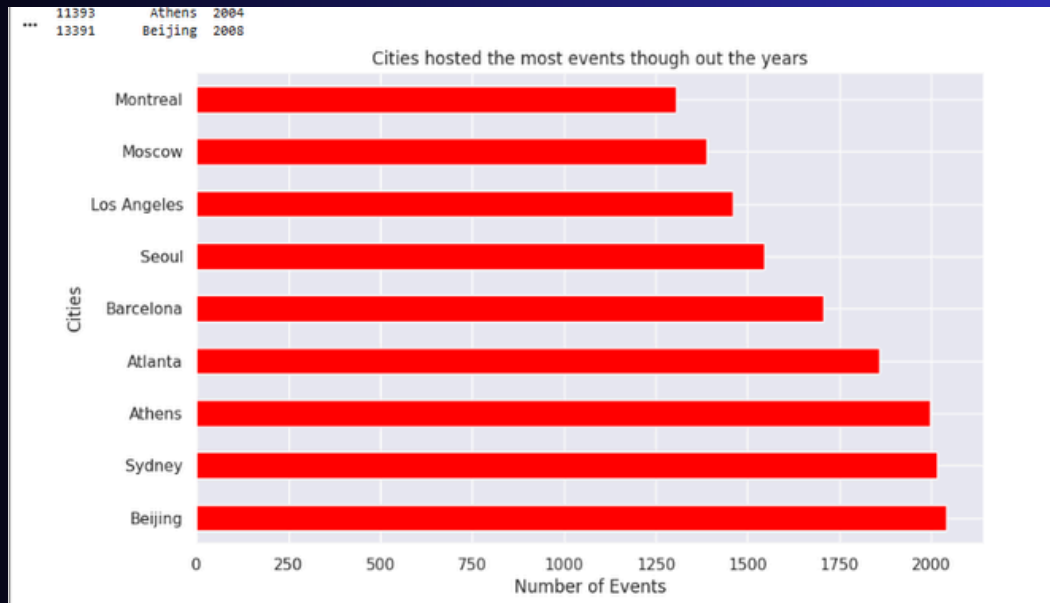
DATA VISUALIZATION

CITIES THAT HOSTED OLYMPICS

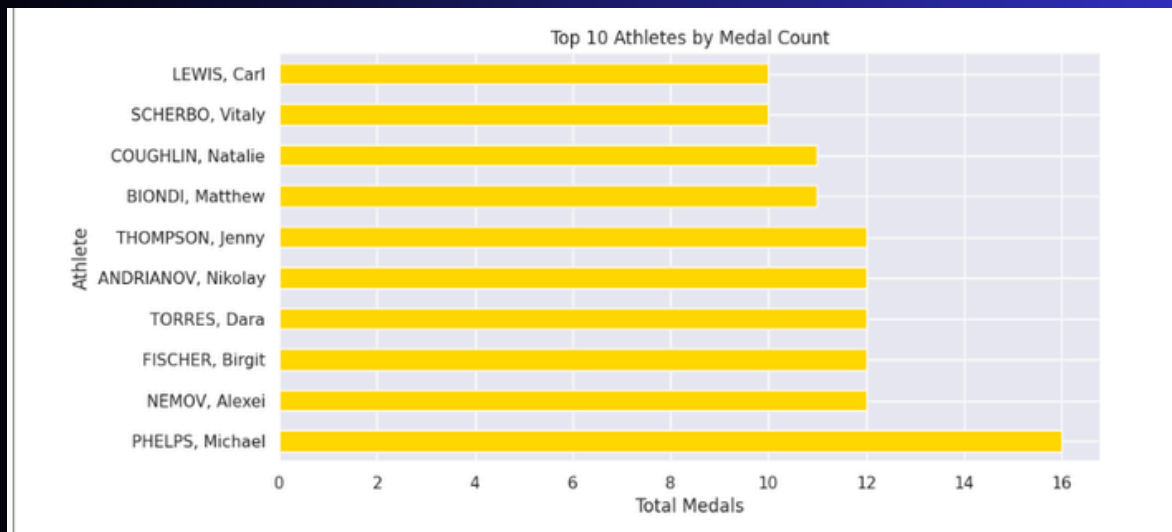
```
=====
CITIES THAT HOSTED THE OLYMPICS (1976-2008)
=====
```

	City	Year
0	Montreal	1976
1422	Moscow	1980
2809	Los Angeles	1984
4268	Seoul	1988
5814	Barcelona	1992
7519	Atlanta	1996
9378	Sydney	2000
11393	Athens	2004
13391	Beijing	2008

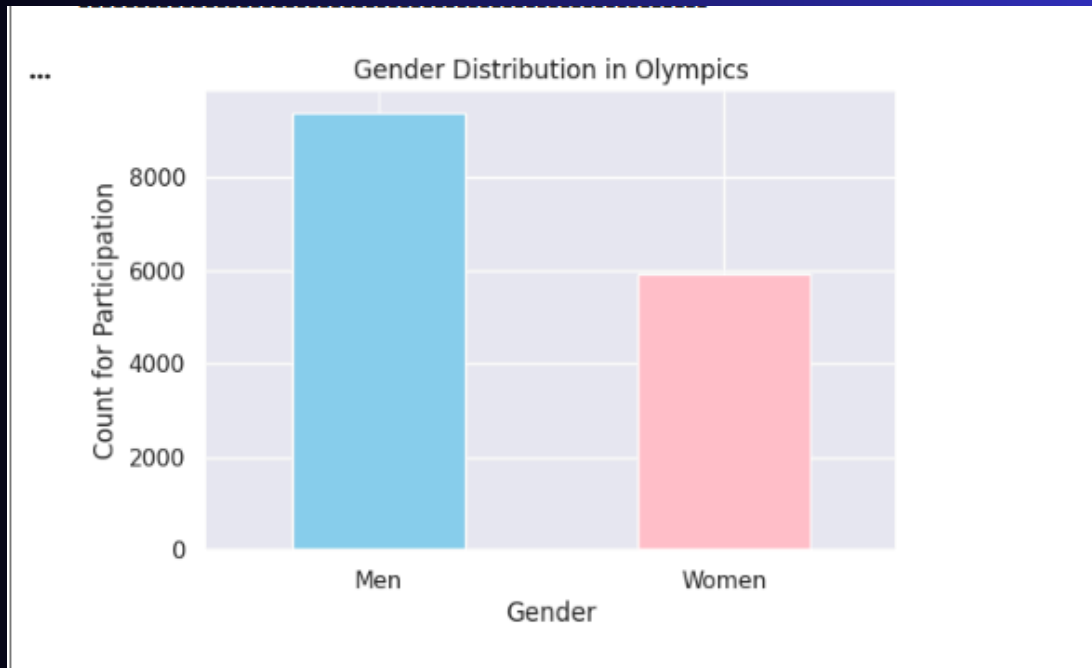
WHICH CITIES HOSTED THE MOST EVENTS



WHICH ATHLETE WON THE MOST MEDALS

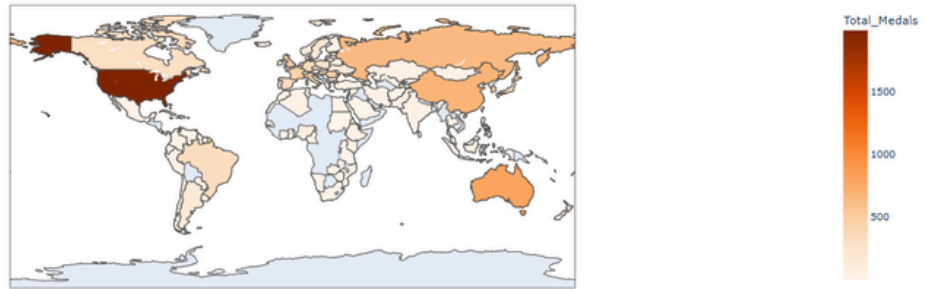


GENDER DISTRIBUTION

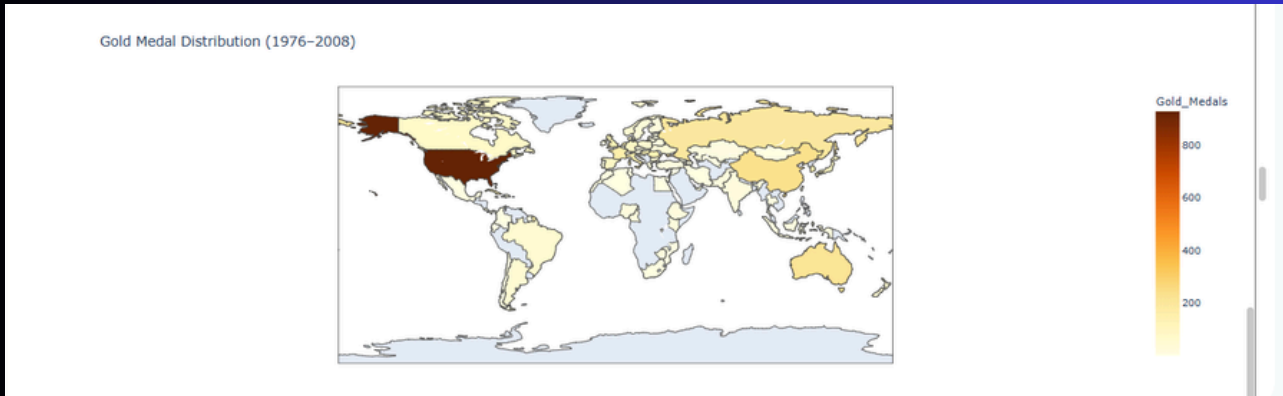


TOTAL MEDAL PER COUNTRY

Total Olympic Medals by Country (1976–2008)



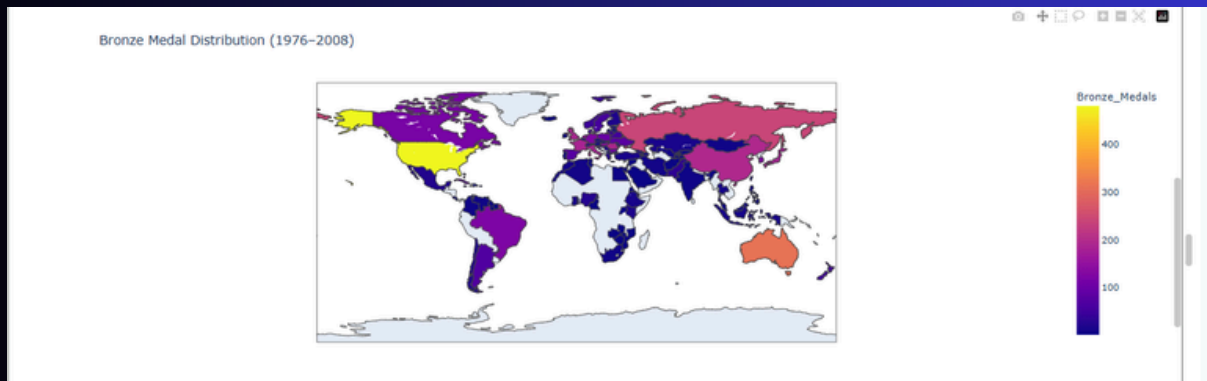
GOLD MEDALS MAP



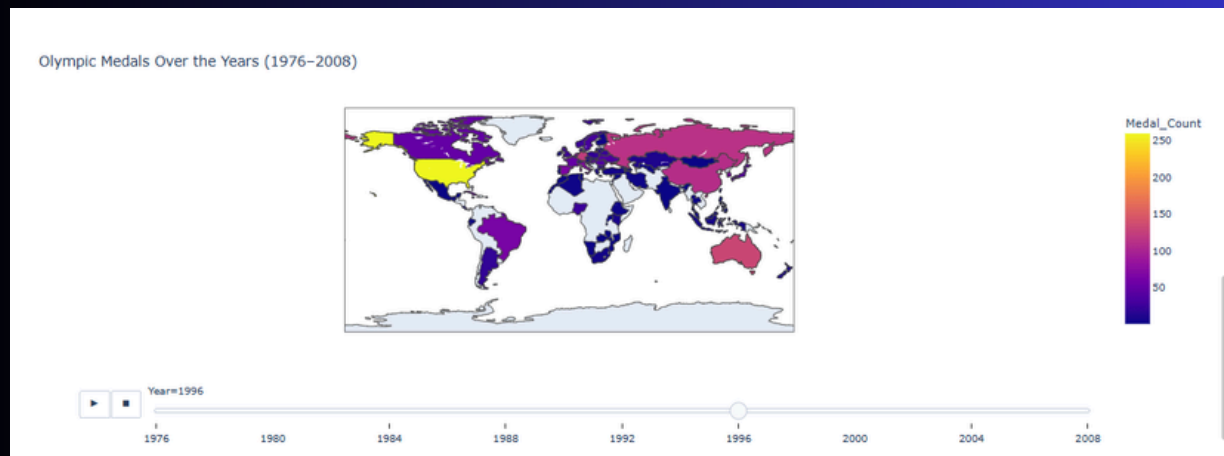
SILVER MEDALS MAP



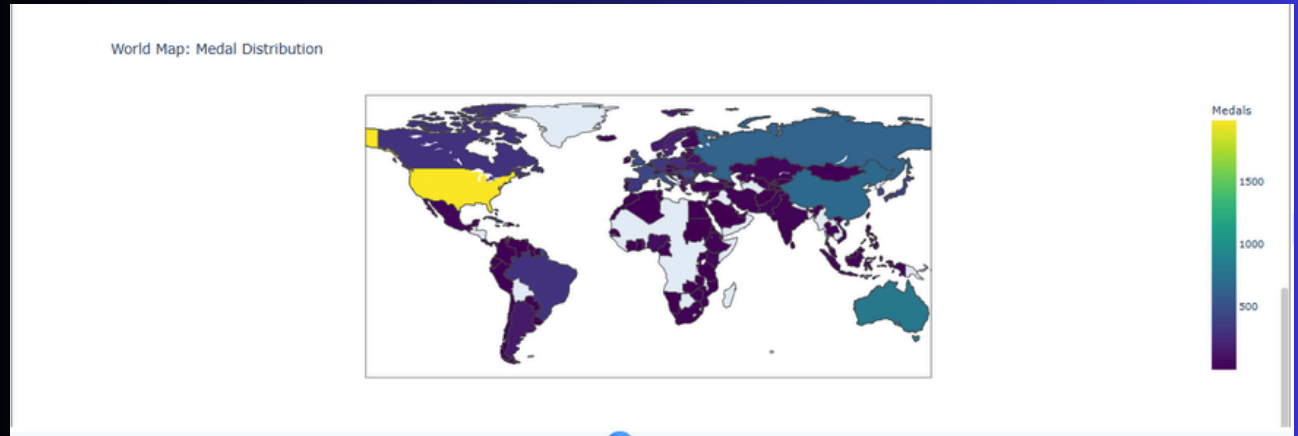
BRONZE MEDALS MAP



MEDALS OVER YEARS



TREND OF MEDALS OVER THE YEARS



KEY INSIGHT

```
*****
***  KEY INSIGHTS FROM OLYMPICS DATA (1976-2008)
*****

1  COUNTRY PERFORMANCE
- United States secured the highest medal count (1992 medals).
- A small number of countries dominate the majority of medals.

2  GENDER PARTICIPATION
- Men athletes have participated more in the tournament.
- Female participation, although lower, shows steady improvement.

3  ATHLETE PERFORMANCE
- PHELPS, Michael is the most decorated athlete with 16 medals.
- Very few athletes win medals consistently across multiple years.

4  HOSTING CITY OBSERVATIONS
- Beijing hosted the most Olympic events (2042 times).
- Hosting duties rotate widely among global cities.

5  YEARLY MEDAL TREND
- Highest number of medals were awarded in 2008 (2042 medals).
- Medal counts show fluctuations based on participation and event count.

6  GEOGRAPHICAL DISTRIBUTION
- North America, Europe, and parts of Asia lead the medal tally.
- Developing nations show lower medal counts due to resource and training gaps.

*****
      ✓ END OF KEY INSIGHTS
*****
```